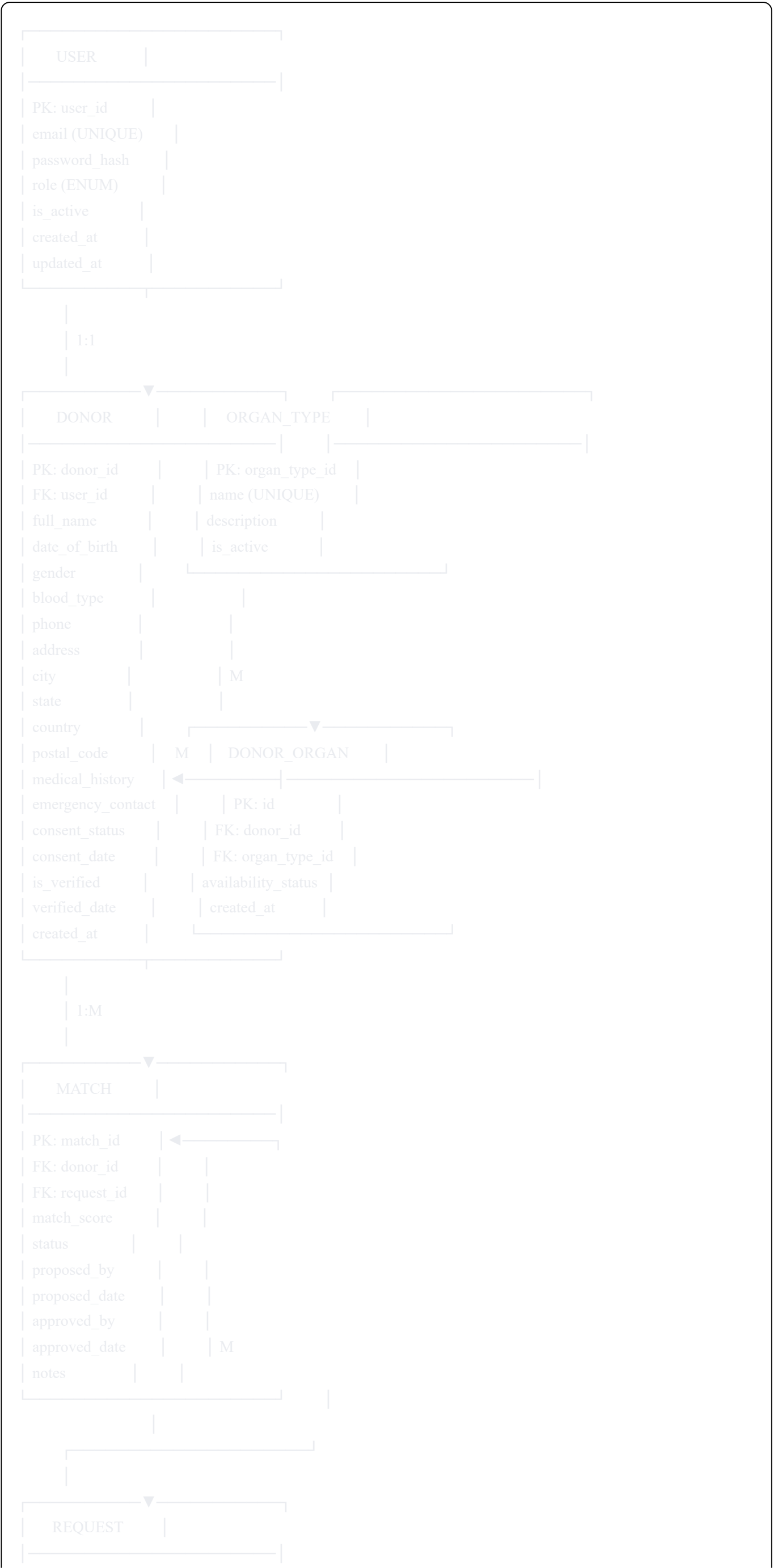


Database Design - LifeLink Organ Donation System

Entity Relationship Diagram (ERD)

ER Diagram (Text Representation)



PK: request_id	
FK: user_id	
FK: organ_type_id	
patient_name	
patient_age	
patient_blood_type	
urgency_level	
hospital_name	
location	
medical_reason	
status	
priority	
created_at	
updated_at	

NOTIFICATION	
PK: notification_id	
FK: user_id	
title	
message	
type	
is_read	
created_at	

AUDIT_LOG	
PK: log_id	
FK: user_id	
action	
entity_type	
entity_id	
old_value	
new_value	
ip_address	
timestamp	

Database Schema (MySQL)

1. USER Table

```
sql

CREATE TABLE user (
  user_id BIGINT AUTO_INCREMENT PRIMARY KEY,
  email VARCHAR(255) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  role ENUM('DONOR', 'HOSPITAL', 'ADMIN') NOT NULL,
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  INDEX idx_email (email),
  INDEX idx_role (role)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Purpose: Central authentication and user management
Relationships: 1:1 with Donor (if role=DONOR), 1:M with Request, Notification, AuditLog

2. DONOR Table

```
sql

CREATE TABLE donor (
  donor_id BIGINT AUTO_INCREMENT PRIMARY KEY,
  user_id BIGINT NOT NULL UNIQUE,
  full_name VARCHAR(255) NOT NULL,
  date_of_birth DATE NOT NULL,
  gender ENUM('MALE', 'FEMALE', 'OTHER') NOT NULL,
  blood_type ENUM('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-') NOT NULL,
  phone VARCHAR(20) NOT NULL,
  address TEXT NOT NULL,
  city VARCHAR(100) NOT NULL,
  state VARCHAR(100) NOT NULL,
  country VARCHAR(100) NOT NULL,
  postal_code VARCHAR(20) NOT NULL,
  medical_history TEXT,
  emergency_contact VARCHAR(255),
  consent_status ENUM('PENDING', 'APPROVED', 'REVOKED') DEFAULT 'PENDING',
  consent_date TIMESTAMP NULL,
  is_verified BOOLEAN DEFAULT FALSE,
  verified_date TIMESTAMP NULL,
  verified_by BIGINT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES user(user_id) ON DELETE CASCADE,
  FOREIGN KEY (verified_by) REFERENCES user(user_id) ON DELETE SET NULL,
  INDEX idx_blood_type (blood_type),
  INDEX idx_city (city),
  INDEX idx_consent_status (consent_status),
  INDEX idx_is_verified (is_verified)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Purpose: Store donor personal and medical information
Relationships: M:1 with User, 1:M with DonorOrgan, 1:M with Match

3. ORGAN_TYPE Table

```
sql

CREATE TABLE organ_type (
  organ_type_id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL UNIQUE,
  description TEXT,
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  INDEX idx_name (name)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- Pre-populate with common organs
INSERT INTO organ_type (name, description) VALUES
('Heart', 'Cardiac organ responsible for pumping blood'),
('Liver', 'Organ responsible for detoxification and metabolism'),
('Kidney', 'Organ responsible for filtering blood and producing urine'),
('Lung', 'Respiratory organ for gas exchange'),
('Pancreas', 'Organ producing insulin and digestive enzymes'),
('Cornea', 'Transparent front part of the eye'),
('Skin', 'Largest organ, used for grafts'),
('Bone', 'Skeletal tissue for transplants'),
('Heart Valve', 'Cardiac valve replacement'),
('Intestine', 'Digestive tract organ');
```

Purpose: Maintain list of donatable organs
Relationships: 1:M with DonorOrgan, 1:M with Request

4. DONOR_ORGAN Table

```
sql

CREATE TABLE donor_organ (
  id BIGINT AUTO_INCREMENT PRIMARY KEY,
  donor_id BIGINT NOT NULL,
  organ_type_id INT NOT NULL,
  availability_status ENUM('AVAILABLE', 'RESERVED', 'DONATED', 'UNAVAILABLE') DEFAULT 'AVAILABLE',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (donor_id) REFERENCES donor(donor_id) ON DELETE CASCADE,
  FOREIGN KEY (organ_type_id) REFERENCES organ_type(organ_type_id) ON DELETE CASCADE,
  UNIQUE KEY unique_donor_organ (donor_id, organ_type_id),
  INDEX idx_availability (availability_status),
  INDEX idx_organ_type (organ_type_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Purpose: Track which organs each donor is willing to donate
Relationships: M:1 with Donor, M:1 with OrganType

5. REQUEST Table

```
sql

CREATE TABLE request (
  request_id BIGINT AUTO_INCREMENT PRIMARY KEY,
  user_id BIGINT NOT NULL,
  organ_type_id INT NOT NULL,
  patient_name VARCHAR(255) NOT NULL,
  patient_age INT NOT NULL,
  patient_blood_type ENUM('A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-') NOT NULL,
  urgency_level ENUM('LOW', 'MEDIUM', 'HIGH', 'CRITICAL') NOT NULL,
  hospital_name VARCHAR(255) NOT NULL,
  hospital_location VARCHAR(255) NOT NULL,
  medical_reason TEXT NOT NULL,
  additional_notes TEXT,
  status ENUM('PENDING', 'UNDER_REVIEW', 'APPROVED', 'MATCHED', 'COMPLETED', 'REJECTED', 'CANCELLED'),
  priority INT DEFAULT 0,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  reviewed_by BIGINT NULL,
  reviewed_at TIMESTAMP NULL,
  FOREIGN KEY (user_id) REFERENCES user(user_id) ON DELETE CASCADE,
  FOREIGN KEY (organ_type_id) REFERENCES organ_type(organ_type_id) ON DELETE CASCADE,
  FOREIGN KEY (reviewed_by) REFERENCES user(user_id) ON DELETE SET NULL,
  INDEX idx_status (status),
  INDEX idx_urgency (urgency_level),
  INDEX idx_organ_type (organ_type_id),
  INDEX idx_blood_type (patient_blood_type),
  INDEX idx_created_at (created_at)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Purpose: Store organ requests from hospitals/recipients
Relationships: M:1 with User, M:1 with OrganType, 1:M with Match

6. MATCH Table

```
sql
```

```
CREATE TABLE `match` (  
  match_id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  donor_id BIGINT NOT NULL,  
  request_id BIGINT NOT NULL,  
  match_score DECIMAL(5,2) DEFAULT 0.00,  
  compatibility_notes TEXT,  
  status ENUM('PROPOSED', 'ACCEPTED', 'REJECTED', 'COMPLETED') DEFAULT 'PROPOSED',  
  proposed_by BIGINT NOT NULL,  
  proposed_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  approved_by BIGINT NULL,  
  approved_date TIMESTAMP NULL,  
  notes TEXT,  
  FOREIGN KEY (donor_id) REFERENCES donor(donor_id) ON DELETE CASCADE,  
  FOREIGN KEY (request_id) REFERENCES request(request_id) ON DELETE CASCADE,  
  FOREIGN KEY (proposed_by) REFERENCES user(user_id) ON DELETE CASCADE,  
  FOREIGN KEY (approved_by) REFERENCES user(user_id) ON DELETE SET NULL,  
  INDEX idx_status (status),  
  INDEX idx_donor (donor_id),  
  INDEX idx_request (request_id)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Purpose: Track proposed donor-recipient matches
Relationships: M:1 with Donor, M:1 with Request

7. NOTIFICATION Table

```
sql  
  
CREATE TABLE notification (  
  notification_id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  user_id BIGINT NOT NULL,  
  title VARCHAR(255) NOT NULL,  
  message TEXT NOT NULL,  
  type ENUM('INFO', 'SUCCESS', 'WARNING', 'ERROR') DEFAULT 'INFO',  
  is_read BOOLEAN DEFAULT FALSE,  
  related_entity_type VARCHAR(50),  
  related_entity_id BIGINT,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (user_id) REFERENCES user(user_id) ON DELETE CASCADE,  
  INDEX idx_user_id (user_id),  
  INDEX idx_is_read (is_read),  
  INDEX idx_created_at (created_at)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Purpose: Store in-app notifications for users
Relationships: M:1 with User

8. AUDIT_LOG Table

```
sql
```

```
CREATE TABLE audit_log (  
  log_id BIGINT AUTO_INCREMENT PRIMARY KEY,  
  user_id BIGINT NULL,  
  action VARCHAR(100) NOT NULL,  
  entity_type VARCHAR(50) NOT NULL,  
  entity_id BIGINT NOT NULL,  
  old_value TEXT,  
  new_value TEXT,  
  ip_address VARCHAR(45),  
  user_agent VARCHAR(255),  
  timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (user_id) REFERENCES user(user_id) ON DELETE SET NULL,  
  INDEX idx_user_id (user_id),  
  INDEX idx_entity (entity_type, entity_id),  
  INDEX idx_timestamp (timestamp),  
  INDEX idx_action (action)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Purpose: Audit trail for compliance and security

Relationships: M:1 with User

Normalization

Normal Forms Achieved: 3NF (Third Normal Form)

- 1NF:** All tables have atomic values, no repeating groups
- 2NF:** All non-key attributes fully dependent on primary key
- 3NF:** No transitive dependencies

Indexing Strategy

Primary Indexes:

- All primary keys (user_id, donor_id, request_id, etc.)

Foreign Key Indexes:

- All foreign keys for join performance

Query Optimization Indexes:

- user.email - Login queries
- donor.blood_type - Donor search
- donor.city - Location-based search
- request.status - Status filtering
- request.urgency_level - Priority sorting
- notification.is_read - Unread notification count

Data Dictionary

DONOR Table Fields

Field Name	Data Type	Constraints	Description
donor_id	BIGINT	PK, AUTO_INCREMENT	Unique donor identifier
user_id	BIGINT	FK, NOT NULL, UNIQUE	Reference to user account
full_name	VARCHAR(255)	NOT NULL	Legal full name
date_of_birth	DATE	NOT NULL	Birth date for age calculation

Field Name	Data Type	Constraints	Description
gender	ENUM	NOT NULL	Gender (MALE, FEMALE, OTHER)
blood_type	ENUM	NOT NULL	ABO blood type
phone	VARCHAR(20)	NOT NULL	Contact phone number
address	TEXT	NOT NULL	Street address
city	VARCHAR(100)	NOT NULL	City of residence
state	VARCHAR(100)	NOT NULL	State/province
country	VARCHAR(100)	NOT NULL	Country
postal_code	VARCHAR(20)	NOT NULL	ZIP/postal code
medical_history	TEXT	NULL	Medical conditions, allergies
emergency_contact	VARCHAR(255)	NULL	Emergency contact details
consent_status	ENUM	DEFAULT 'PENDING'	Consent approval status
is_verified	BOOLEAN	DEFAULT FALSE	Admin verification flag

REQUEST Table Fields

Field Name	Data Type	Constraints	Description
request_id	BIGINT	PK, AUTO_INCREMENT	Unique request identifier
user_id	BIGINT	FK, NOT NULL	Hospital user submitting
organ_type_id	INT	FK, NOT NULL	Requested organ type
patient_name	VARCHAR(255)	NOT NULL	Patient name (can be anonymized)
patient_blood_type	ENUM	NOT NULL	Patient's blood type
urgency_level	ENUM	NOT NULL	Medical urgency (LOW to CRITICAL)
status	ENUM	DEFAULT 'PENDING'	Request workflow status

Sample Data

Sample Users

sql

```
INSERT INTO user (email, password_hash, role) VALUES
('john.donor@email.com', '$2a$10$...', 'DONOR'),
('hospital1@medical.com', '$2a$10$...', 'HOSPITAL'),
('admin@lifelink.org', '$2a$10$...', 'ADMIN');
```

Sample Donor

sql

```
INSERT INTO donor (user_id, full_name, date_of_birth, gender, blood_type, phone, address, city, state, country, postal_code) VALUES
(1, 'John Doe', '1990-05-15', 'MALE', 'O+', '+1234567890', '123 Main St', 'New York', 'NY', 'USA', '10001');
```

Sample Donor Organs

sql

```
INSERT INTO donor_organ (donor_id, organ_type_id, availability_status) VALUES
(1, 3, 'AVAILABLE'), -- Kidney
(1, 2, 'AVAILABLE'); -- Liver
```

Sample Request

```
sql
INSERT INTO request (user_id, organ_type_id, patient_name, patient_age, patient_blood_type, urgency_level, hospital_name, location, medical_history, status)
(2, 3, 'Patient A', 45, 'O+', 'HIGH', 'City Hospital', 'New York, NY', 'Chronic kidney disease', 'PENDING');
```

Backup & Maintenance

Backup Strategy:

- Daily automated full backups at 2 AM
- Hourly incremental backups
- 30-day retention period
- Off-site backup storage

Maintenance Tasks:

- Weekly OPTIMIZE TABLE commands
- Monthly ANALYZE TABLE for statistics
- Quarterly index review and optimization
- Annual archive of old audit logs (>2 years)

Security Measures

1. Encryption:

- Passwords: BCrypt hashing
- Sensitive fields: AES-256 encryption at application layer
- Transport: TLS 1.3

2. Access Control:

- Database user per environment
- Least privilege principle
- No direct database access for application users

3. SQL Injection Prevention:

- Prepared statements only
- Input validation
- ORM (Hibernate) usage

4. Audit Trail:

- All data modifications logged
- Retention: 7 years for compliance

Database Version: MySQL 8.0+

Character Set: utf8mb4

Collation: utf8mb4_unicode_ci

Storage Engine: InnoDB (ACID compliance)